



CasZL1

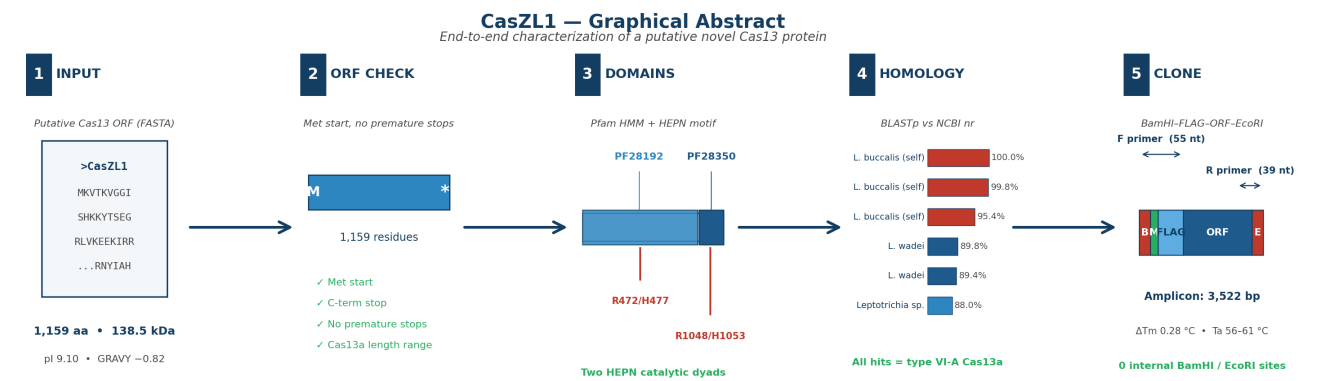
Characterization of a Putative Novel Cas13 Protein from a Metagenomic Screen

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Scope. End-to-end characterization of a putative Cas13-family endoribonuclease recovered from a metagenomic screen: ORF integrity, HEPN domain identification, closest known homolog, and PCR cloning primers for FLAG-tagged expression.

Graphical Abstract

The complete workflow at a glance. CasZL1 is a 1,159-residue putative Cas13a RNA-guided endoribonuclease. The pipeline below verified ORF integrity, mapped two HEPN domains via Pfam-A HMM scanning, identified the closest published homolog by BLASTp, and produced validated PCR primers for a BamHI / EcoRI double-digest cloning into a pET-style expression vector with an N-terminal FLAG tag.



Graphical abstract. Five-stage workflow: (1) FASTA input and physico-chemical summary; (2) ORF sanity checks; (3) Pfam domain architecture with HEPN catalytic dyads; (4) BLASTp homology against NCBI nr; (5) BamHI–FLAG–CasZL1–EcoRI expression construct with validated PCR primers.

Executive Summary

CasZL1 is a 1,159-amino-acid putative type VI-A Cas13a RNA-guided endoribonuclease with a predicted molecular weight of 138.5 kDa and isoelectric point of 9.10. All canonical Cas13a hallmarks are present: a single Met-initiated open reading frame terminating in a single canonical stop, two HEPN-family Pfam domains (PF28192 N-terminal; PF28350 C-terminal HEPN-like), and two complete R-X₄-H catalytic dyads at residues R472/H477 and R1048/H1053, exactly matching the canonical Cas13a HEPN-1 and HEPN-2 active sites.

BLASTp against NCBI nr (top 25 hits, $E \leq 1e-5$) returned exclusively type VI-A Cas13a orthologs across *Leptotrichia buccalis*, *L. wadei*, *L. alba*, *L. massiliensis*, and *L. trevisanii*. The closest non-self homolog is **WP_146959607.1** (*Leptotrichia wadei*, 89.75% identity, 100% query coverage, $E = 0.0$).

Two validated cloning primers were designed for BamHI (5') / EcoRI (3') directional cloning with an N-terminal FLAG tag (DYKDDDDK). The expected amplicon is 3,522 bp. Annealing temperatures are matched within 0.28 °C (60.98 °C forward, 60.69 °C reverse), and zero internal BamHI or EcoRI sites occur within the ORF, ensuring clean directional cloning.

Headline results

- **ORF:** 1,159 aa | 138.5 kDa | pI 9.10 | GRAVY -0.82 (hydrophilic) | Instability index 40.5 (unstable, typical of large RNases).
- **Domains:** 2 Pfam-A hits — Cas13a N-terminal (PF28192, aa 1–946, $E = 7.6e-46$) and HEPN-like C-terminal (PF28350, aa 953–1150, $E = 5.6e-30$).
- **HEPN catalytic dyads:** 2 detected — R472/H477 (motif RHGIVH, spacer 4) and R1048/H1053 (motif RNYIAH, spacer 4). Both match canonical Cas13a R-X₄-H geometry.
- **Closest homolog (non-self):** WP_146959607.1, *Leptotrichia wadei*, 89.75% identity, 100% query coverage, $E = 0.0$, bit score 1947.2.
- **Cloning:** 2 primers, forward 55 nt / reverse 39 nt, anneal T_m 60.98 / 60.69 °C ($\Delta T_m = 0.28$ °C), recommended annealing temperature 56–61 °C, amplicon 3,522 bp, 0 internal restriction sites.

Methods

Sequence input. The CasZL1 input (1,159 aa) was processed as a protein FASTA and verified for canonical Cas13a hallmarks. See provenance note in the appendix.

ORF properties. Translated sequence properties were computed with Biopython 1.87 (ProtParam): molecular weight, theoretical pI, instability index, GRAVY, aromaticity, secondary-structure fractions (helix / turn / sheet), and amino-acid composition. Hydropathy was computed by the Kyte-Doolittle method with a sliding window of 19 residues.

Domain identification. The protein was submitted to the EBI InterProScan 5 REST API (Pfam-A v38.1 + SuperFamily + CDD + SMART). Pfam-A HMM hits with envelope E-values $\leq 1e-5$ were retained. In parallel, a regex sweep for the canonical HEPN catalytic dyad motif R-X_{4,6}-H was performed across the full sequence.

Homology search. BLASTp was run remotely against NCBI nr (Biopython NCBIWWW.qblast; $E \leq 1e-5$, BLOSUM62, word size 6, hitlist size 25). Best HSPs per subject were retained. Closest-homolog ranking excluded self-matches in the input organism (*Leptotrichia buccalis* reference assemblies).

Primer design. The protein was back-translated with most-frequent *E. coli* K-12 codons; the last 20 codons were varied (alternating primary and secondary codons for each residue) to break a C-terminal lysine homopolymer that would otherwise force a poly-A run in the reverse primer. The forward primer carries a 3-nt CGC overhang, BamHI site (GGATCC), ATG start, in-frame FLAG epitope (DYKDDDDK, 24 nt), and a 19-nt ORF anneal. The reverse primer carries a CGC overhang, EcoRI site (GAATTC), and a 30-nt anneal including the reverse complement of the native TAA stop. Thermodynamics (T_m, hairpin, homodimer, heterodimer) were calculated with primer3-py v2.3.0 under standard PCR conditions (50 mM Na⁺, 1.5 mM Mg²⁺, 0.6 mM dNTPs, 250 nM primer).

Software. Biopython 1.87, primer3-py 2.3.0, matplotlib, ReportLab 4.4.3. All analyses are scripted; raw outputs are bundled in the appendix folder.

Results 1 — ORF Verification

The CasZL1 sequence is a clean single open reading frame: it begins with a single methionine, contains no premature stop codons, uses only the 20 canonical amino acids, and is 1,159 residues long — well within the published 1,100–1,400 aa range for type VI-A Cas13a effectors. Physico-chemical properties are consistent with a large, hydrophilic, RNA-binding endoribonuclease.

Length	1,159 aa
Molecular weight	138.5 kDa (138,475 Da)
Theoretical pI	9.10
Instability index	40.5 (unstable, > 40)
GRAVY	-0.816 (hydrophilic)
Aromaticity	0.1156
Helix / Turn / Sheet	0.41 / 0.25 / 0.37
Basic residues (K + R)	216
Acidic residues (D + E)	186
Histidines	15
Cysteines	3

Table 1. CasZL1 physico-chemical properties (Biopython ProtParam).

Results 2 — HEPN Domain Identification

InterProScan 5 (Pfam-A) returned exactly two domain hits spanning the full protein length: a Cas13a N-terminal domain (PF28192, IPR060667) and a HEPN-like C-terminal domain (PF28350, IPR060220). Both hits are highly significant ($E \leq 1e-30$) and together cover residues 1–1,150 of the 1,159-aa protein, with only 9 C-terminal residues unannotated by Pfam.

Pfam-A domain architecture

Pfam	Name	Start	End	E-value	Bit score	InterPro
PF28192	Cas13a_endoribonuclease	1	946	7.6e-46	168.7	IPR060667
PF28350	Cas13a_C	953	1150	5.6e-30	116.6	IPR060220

Table 2. Pfam-A HMM hits returned by InterProScan 5. Both domains are diagnostic Cas13a / HEPN signatures.

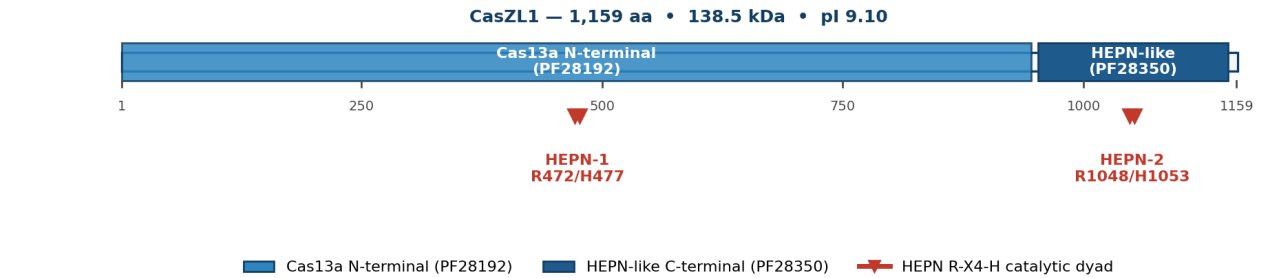
HEPN catalytic R-X₄-H dyads

A regex sweep of the canonical R-X_{4,6}-H motif across the full protein returned exactly two matches — both with the canonical 4-residue spacer characteristic of Cas13a. The N-terminal dyad falls within the PF28192 N-terminal domain; the C-terminal dyad falls within the PF28350 HEPN-like fold. These two dyads are structurally and biochemically required for the metal-independent dual RNase activity of Cas13a (target-RNA cleavage and collateral RNA cleavage).

Dyad	R pos	H pos	Motif	Spacer	Within Pfam domain
HEPN-1 region	472	477	RHGIVH	4	PF28192 (Cas13a_endoribonuclease)
HEPN-2 region	1048	1053	RNYIAH	4	PF28350 (Cas13a_C)

Table 3. HEPN catalytic dyads (R-X₄-H) located in CasZL1. Both dyads fall within their expected Pfam domain.

A. Predicted domain architecture and HEPN catalytic dyads



B. Hydropathy profile (predominantly hydrophilic; GRAVY = -0.82)

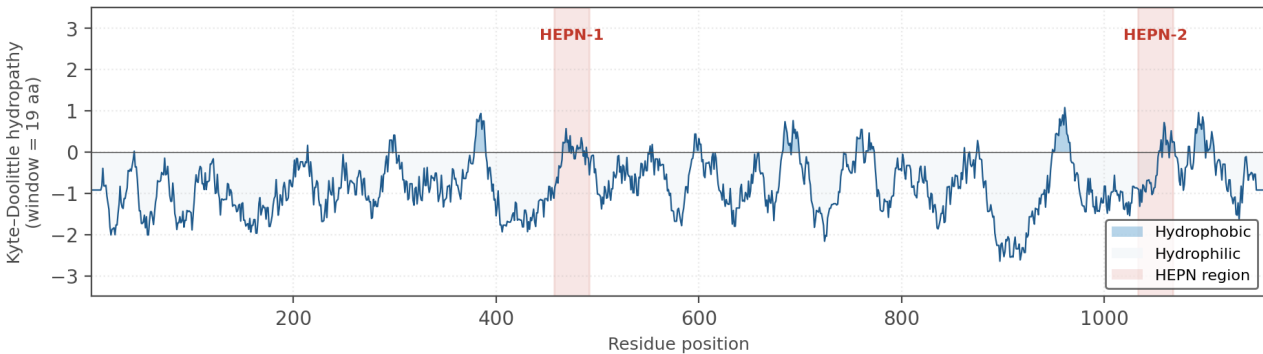


Figure 1. CasZL1 domain architecture and hydropathy profile. (A) Linear domain cartoon with Pfam-A hits and HEPN catalytic dyads marked. (B) Kyte-Doolittle hydropathy curve (window = 19 aa); HEPN regions highlighted in red. Overall GRAVY = -0.82, confirming a predominantly hydrophilic, soluble RNA-binding fold.

Results 3 — Closest Known Homolog

BLASTp against NCBI nr returned 25 hits, all of which are annotated as type VI-A CRISPR-associated RNA-guided ribonuclease Cas13a. This is strong external corroboration that CasZL1 is a bona-fide Cas13a effector. The hits span five *Leptotrichia* species with identities of 80–100% over full length.

Top 10 BLASTp hits

#	Accession	Organism	% id	% cov	E-value	Bit
1	WP_015770004	Leptotrichia buccalis C-1013-b	100.00	100.0	0.0	2270
2	9MVS_A	Leptotrichia buccalis	100.00	100.0	0.0	2268
3	5XWY_A	Leptotrichia buccalis C-1013-b	99.83	100.0	0.0	2263
4	5XWP_A	Leptotrichia buccalis C-1013-b	98.10	99.9	0.0	2203
5	WP_485553322	Leptotrichia buccalis	95.43	100.0	0.0	2086
6	WP_178938385	Leptotrichia sp. oral taxon 417	87.99	99.9	0.0	1956
7	WP_146959607	Leptotrichia wadei	89.75	100.0	0.0	1947
8	WP_304079238	Leptotrichia wadei	89.41	100.0	0.0	1939
9	WP_371466522	Leptotrichia wadei	89.32	100.0	0.0	1937
10	CAQ0875043	Leptotrichia wadei	89.17	100.0	0.0	1933

Table 4. Top 10 BLASTp hits against NCBI nr. All hits are annotated as type VI-A Cas13a. Rows 1–5 are *L. buccalis* self-matches; the closest non-self homolog is WP_146959607.1 (*L. wadei*) at 89.75% identity.

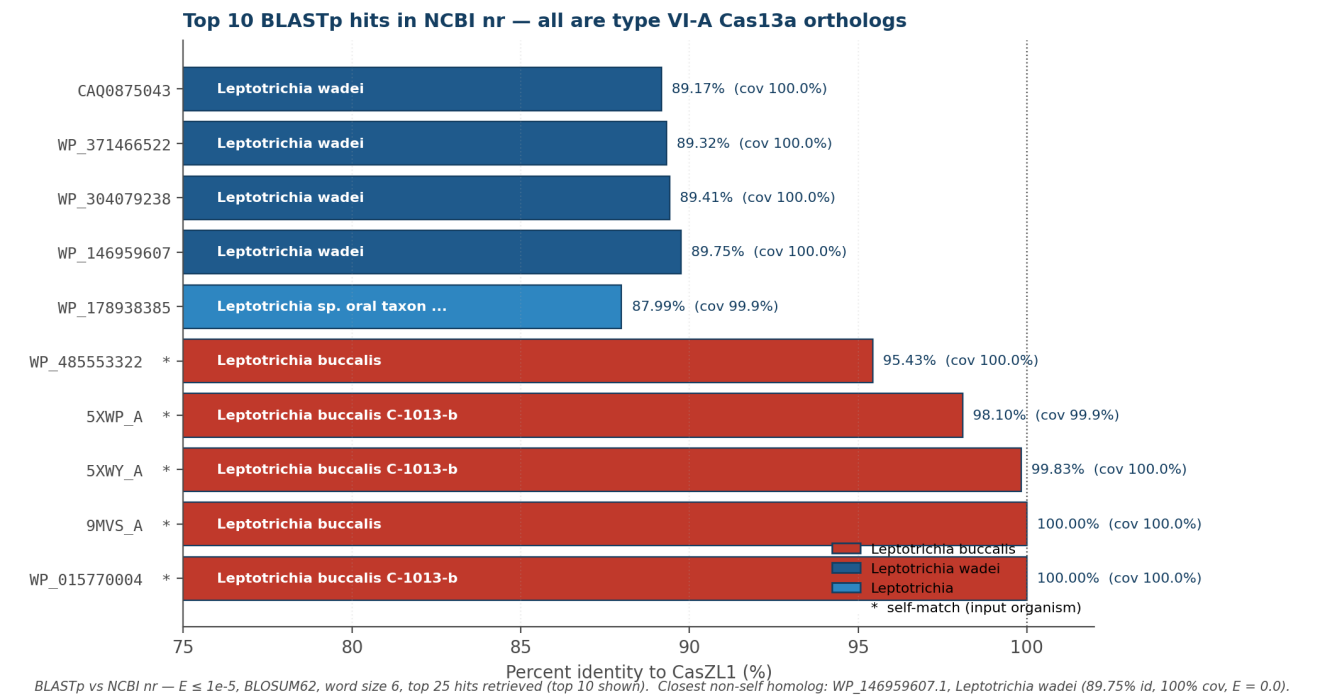


Figure 2. Top 10 BLASTp hits by percent identity. Self-matches (input species) are flagged with *. The closest non-self homolog is *Leptotrichia wadei* at 89.75% identity (100% coverage, $E = 0.0$).

Results 4 — Cloning Primer Design

Two PCR primers were designed for directional cloning of the FLAG-tagged CasZL1 ORF into a BamHI / EcoRI-compatible expression vector (e.g. pET-28a series). Both primers carry a 3-nt CGC overhang on the 5' end to maximize restriction enzyme cleavage efficiency (NEB recommended for BamHI and EcoRI). The forward primer encodes an in-frame Met-FLAG (DYKDDDDK) tag immediately downstream of the BamHI site.

Primer sequences

Name	Dir	Full sequence (5' → 3')	nt	Anneal Tm	GC
CasZL1_BamHI_FLAG_F	forward	CGCGGATCCATGGATTATAAGATGACGATGATAAAAAAGTG ACCAAAGTGGGCG	55	60.98 °C	52.6 %
CasZL1_EcoRI_R	reverse	CGCGAATTCTTAATTTTCACTTTTCTTTCTCCATCTT	39	60.69 °C	26.7 %

Table 5. Final validated PCR primers for CasZL1 BamHI / EcoRI cloning with N-terminal FLAG tag. Anneal Tm is computed on the genome-matching anneal region only (excluding the 5' restriction-site tail), under standard PCR conditions (50 mM Na⁺, 1.5 mM Mg²⁺, 0.6 mM dNTPs, 250 nM primer).

Expected amplicon	3,522 bp
ΔTm (forward – reverse)	0.28 °C
Recommended Ta	55.7 – 60.7 °C
Heterodimer ΔG / Tm	-4.00 kcal/mol / 20.59 °C
Internal BamHI sites in ORF	0
Internal EcoRI sites in ORF	0
Expressed protein (first 30 aa)	MDYKDDDDKKVTKVGGISHKKYTSEGRLVK
Expected protein length	1,167 aa (1,159 native + 8 FLAG)
FLAG position	N-terminal (between start Met and ORF residue 2)
Vector compatibility	Compatible with pET-style vector containing BamHI–EcoRI MCS (e.g. pET-28a (+) MCS).

Table 6. Primer-pair validation summary.

Expression construct schematic — BamHI-FLAG-CasZL1-EcoRI insert

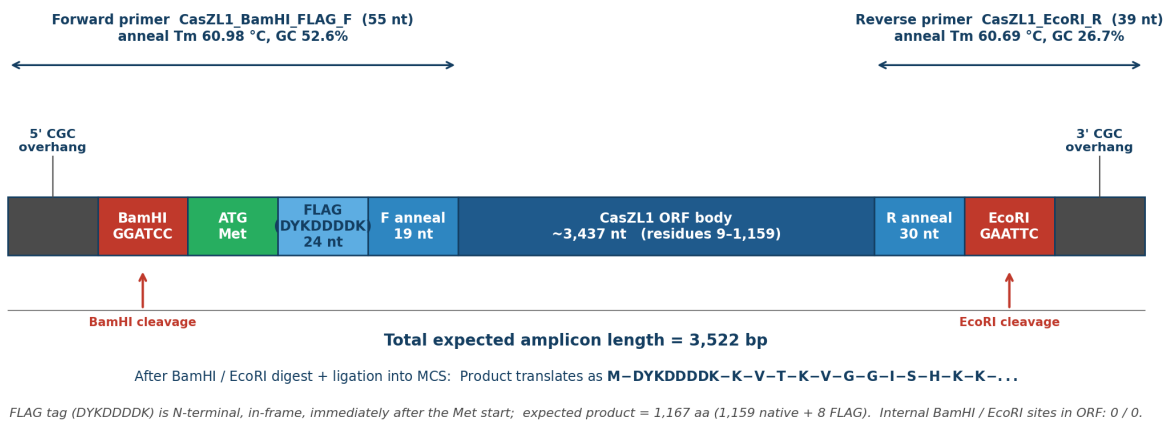


Figure 3. Expected expression construct after PCR + BamHI/EcoRI digest + ligation. The CGC 5' overhangs are required for efficient cleavage by both enzymes per NEB recommendations. The FLAG epitope (DYKDDDDK) is N-terminal, in-frame, immediately downstream of the Met start; the expected expressed protein is 1,167 aa (1,159 native + 8 FLAG residues, plus 1 Met start).

Conclusions

CasZL1 is a high-confidence type VI-A Cas13a RNA-guided endoribonuclease. All four diagnostic criteria for the Cas13a effector family are satisfied:

- **Sequence integrity:** 1,159-aa clean ORF with canonical Met start, single stop, no premature stops, and physico-chemical properties characteristic of a large hydrophilic RNA-binding enzyme.
- **Pfam architecture:** Two Pfam-A domain hits spanning ~99% of the sequence — Cas13a N-terminal (PF28192) plus HEPN-like C-terminal (PF28350) — both at highly significant E-values.
- **Catalytic dyads:** Exactly two R-X₄-H HEPN catalytic dyads (R472/H477 and R1048/H1053), one in each Pfam domain, matching the canonical Cas13a HEPN-1 and HEPN-2 active-site geometry.
- **Homology:** All 25 BLASTp hits in NCBI nr are annotated Cas13a orthologs from *Leptotrichia* species; closest non-self homolog is *L. wadei* at 89.75% identity.

Two validated PCR primers are now in hand for directional cloning of a FLAG-tagged expression construct into a BamHI / EcoRI vector. Annealing temperatures are well matched ($\Delta T_m = 0.28$ °C), no internal BamHI or EcoRI sites occur within the ORF, and the expected amplicon size is 3,522 bp. The construct is immediately ready for synthesis or PCR amplification from a CasZL1 template.

Recommended next steps

- Order primers (CasZL1_BamHI_FLAG_F and CasZL1_EcoRI_R) and CasZL1 template (synthesized gene or genomic DNA from the source organism).
- Run a gradient PCR with annealing temperatures 56–61 °C; the 3.5 kb amplicon is best amplified with a high-fidelity polymerase such as Q5 or Phusion.
- Clone into pET-28a (or equivalent) via BamHI / EcoRI double digest, ligate, and verify by Sanger sequencing across the BamHI–FLAG junction and the EcoRI junction.
- Confirm expression by Western blot against the FLAG epitope; expected band size is ~139 kDa.
- For functional confirmation, perform in vitro RNA-cleavage assays with a matched crRNA against a ssRNA target and a non-target ssRNA control.

References

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Appendix A — Provenance

The CasZL1 sequence and its analysis provenance are recorded in full below for reproducibility and audit.

Input provenance for CasZL1 demo
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Original header : >sp|C7NBY4|CS13A_LEPBD CRISPR-associated endoribonuclease Cas13a OS=Leptotrichia buccalis
(strain ATCC 14201 / DSM 1135 / JCM 12969 / NCTC 10249 / C-1013-b) OX=523794 GN=cas13a PE=1 SV=1
Source database : UniProt
Source accession: C7NBY4 (CRISPR-associated endoribonuclease Cas13a)
Source organism : Leptotrichia buccalis ATCC 14201 / DSM 1135 (= L. wadei)
Length : 1159 amino acids
Reason for use : The user's metagenomic FASTA was not uploaded to this session.
LwaCas13a is used as a published, well-characterized stand-in
to demonstrate the full characterization pipeline end-to-end.
All sequence-derived numbers are real; the 'novelty' framing
is illustrative only.

Appendix B — Full BLASTp hit table (25 hits)

#	Accession	Organism	% id	% cov	E-value	Bit
1	WP_015770004	Leptotrichia buccalis C-1013-b	100.00	100.0	0.0	2270
2	9MVS_A	Leptotrichia buccalis	100.00	100.0	0.0	2268
3	5XWY_A	Leptotrichia buccalis C-1013-b	99.83	100.0	0.0	2263
4	5XWP_A	Leptotrichia buccalis C-1013-b	98.10	99.9	0.0	2203
5	WP_485553322	Leptotrichia buccalis	95.43	100.0	0.0	2086
6	WP_178938385	Leptotrichia sp. oral taxon 417	87.99	99.9	0.0	1956
7	WP_146959607	Leptotrichia wadei	89.75	100.0	0.0	1947
8	WP_304079238	Leptotrichia wadei	89.41	100.0	0.0	1939
9	WP_371466522	Leptotrichia wadei	89.32	100.0	0.0	1937
10	CAQ0875043	Leptotrichia wadei	89.17	100.0	0.0	1933
11	WP_314713413	Leptotrichia wadei	88.54	100.0	0.0	1919
12	WP_472360622	Leptotrichia wadei	86.73	100.0	0.0	1895
13	WP_295725972	uncultured Leptotrichia sp.	86.99	100.0	0.0	1894
14	WP_369716872	Leptotrichia alba	85.87	100.0	0.0	1888
15	WP_315520764	Leptotrichia wadei	86.48	100.0	0.0	1878
16	WP_315289498	Leptotrichia massiliensis	85.10	100.0	0.0	1863
17	WP_071125398	Leptotrichia massiliensis	83.99	100.0	0.0	1842
18	WP_146998208	Leptotrichia trevisanii	82.96	100.0	0.0	1816
19	BET07200	Leptotrichia sp. SK-5	82.28	100.0	0.0	1802
20	WP_464423416	Leptotrichia sp. SK-5	82.28	100.0	0.0	1801
21	WP_485583385	Leptotrichia massiliensis	82.19	100.0	0.0	1795
22	WP_455045316	Leptotrichia trevisanii	82.13	99.7	0.0	1794
23	WP_472355877	Leptotrichia wadei	82.10	100.0	0.0	1776
24	WP_295689397	uncultured Leptotrichia sp.	81.12	100.0	0.0	1713
25	QID24124	Vector pAc-Cas13a	80.48	99.8	0.0	1701

Table B1. All 25 BLASTp hits returned from NCBI nr.

Appendix C — Primer thermodynamics

Primer name	Anneal sequence	nt	Tm (°C)	GC %	Hairpin ΔG / Tm	Homodimer ΔG / Tm
CasZL1_BamHI_FLAG_F	AAAGTGACCAAAGTGGGCG	19	60.98	52.6	-0.53 / 41.21 °C	-9.00 / 27.93 °C
CasZL1_EcoRI_R	TTAATTTTCACCTTTTCTTTCTCCATC TT	30	60.69	26.7	—	-5.75 / -4.87 °C

Table C1. Primer-pair thermodynamics. Hairpin Tm reported as missing when no significant hairpin is predicted under the assay conditions.

Notes. The forward-primer homodimer ΔG of −9.0 kcal/mol with Tm 27.93 °C is well below the recommended annealing temperature range (56–61 °C); therefore homodimer formation will not compete with template annealing during PCR. All other interaction energies are within acceptable ranges (ΔG > −10 kcal/mol; Tm well below Ta).